

1. Client Requirements

1.1 Client Overview

Client: Boitshoko Digital Media Agency, Rustenburg (Client ID: CLI-129). Industry: Media. The agency operates from a single site and requires a converged, segmented network supporting day-to-day media production, client-facing sales, and back-office finance operations, with resilient internet access and secure address translation between the internal network and the public internet.

1.2 Scenario Interpretation & Assumptions

The brief fixes the client, industry, addressing block, design constraint, assigned challenge and change request — these are treated as fixed and non-negotiable. It does not fix department structure, headcounts, device types, or service locations. Per the module's project guidance, the correct response to that gap is not to seek more information from the named organisation, but to make a reasonable, defensible network-engineering assumption, document it, and design against it consistently. The assumptions below are the ones this design relies on; each is a normal, low-risk judgement call for an office network of this size and industry, and each is revisited.

#	Gap in the brief	Assumption made	Justification
A1	Department structure	Six functional departments: Management, Finance, Creative & Design, Client Services & Sales, IT/Network Operations, Reception	Standard structure for a small-to-medium media agency; each department maps to a distinct traffic/security profile, which motivates VLAN segmentation
A2	Headcount per department	5–20 users per department today (53 total); see Section 4.1	Sized to a single-site agency small enough for one border router and one core switch, in line with the brief's single addressing block and single-site framing
A3	Shared services	Internal file server, web/portal server, email server, grouped in a Servers/DMZ VLAN	Typical minimum service set for any office LAN; isolating them in their own VLAN limits the blast radius of a compromised end-user device
A4	Backup path scope	Backup ISP provides failover for the whole router (and therefore Finance) rather than a second, Finance-only router	Meets the constraint literally (Finance never loses internet) with one fewer device and one fewer point of failure than a dedicated Finance router; documented as a deliberate simplification, not an oversight
A5	Printers / peripherals	Not modelled as separate devices in Milestone 1; treated as end hosts on their owning department's VLAN if added in Packet Tracer	Brief does not require print services or a specific location; adding this scope would exceed "solve the assigned problem", per the lecturer's guidance not to expand scope unprompted

1.3 Functional Requirements (derived from the project brief)

- FR1 — Provide full IP connectivity across all operational departments at the Rustenburg site: Management, Finance, Creative & Design, Client Services & Sales, IT/Network Operations, and Reception.
- FR2 — Provide all departments with access to shared internal servers (file storage, internal web/portal, email).
- FR3 — Provide outbound internet access to every department for cloud media transfer, client communication and web-based tools, via a primary ISP connection.

- FR4 — Guarantee that the Finance department's internet access survives a primary ISP outage, via a backup internet path (Design Constraint, Section 8 of the brief).
- FR5 — Translate all internal private addressing (10.49.0.0/16) to public/outside addressing before traffic leaves the network (Assigned Networking Challenge: NAT).
- FR6 — Absorb a 25% increase in user numbers in every department without renumbering any existing host or subnet (Client Change Request CR5).

1.4 Non-Functional / Technical Requirements

- Implemented and simulated entirely in Cisco Packet Tracer.
- VLAN segmentation per department to control broadcast domains and to apply department-appropriate access policy.
- Inter-VLAN routing via a router-on-a-stick border router (single physical router, 802.1Q sub-interfaces).
- NAT overload (PAT) on the primary ISP path so the whole 10.49.0.0/16 inside network shares a small outside address pool.
- A second, independent outside path (separate ISP, separate outside network) that keeps Finance reachable if the primary path fails.
- A reproducible, testable .pkt file plus a GitHub portfolio documenting requirements, design, configuration, testing and reflection.

1.5 Design Constraint

Backup internet path required for the finance function. Interpreted as: the router must present two distinct outside paths to the internet (primary and backup ISP), with automatic failover, so that the Finance VLAN's default route — and by extension every VLAN routed through the same border router — is not dependent on a single ISP. See Assumption A4.

1.6 Client Change Request — CR5

User numbers grow by 25% — the addressing plan must absorb this without renumbering. Interpreted as: every department subnet in Section 4 is sized using the department's headcount after a 25% increase is already applied, and the full 10.49.0.0/16 block is only lightly used (a single /24) so additional departments, sites, or larger-than-expected growth can be added from unused address space without touching any subnet already in service.